Figure 1.9 The original design of the AutoGPT agent system

AutoGPT became the first example to demonstrate the power of using task planning and iteration with LLM models. From this and in tandem, other agent systems and frameworks exploded into the community using similar planning and task iteration systems. It's generally accepted that planning, iteration, and repetition are the best processes for solving complex and multifaceted goals for an LLM.

However, autonomous agent systems require trust in the agent decision-making process, the guardrails/evaluation system, and the goal definition. Trust is also something that is acquired over time. Our lack of trust stems from our lack of understanding of an autonomous agent's capabilities.

> **Note** Artificial general intelligence (AGI) is a form of intelligence that can learn to accomplish any task a human can. Many practitioners in this new world of AI believe an AGI using autonomous agent systems is an attainable goal.

For this reason, many of the mainstream and production-ready agent tools aren't autonomous. However, they still provide a significant benefit in managing and automating tasks using GPTs (LLMs). Therefore, as our goal in this book is to understand all agent forms, many more practical applications will be driven by non-autonomous agents.

Agents and agent tools are only the top layer of a new software application development paradigm. We'll look at this new paradigm in the next section.

1.4 Peeling back the AI interface

The AI agent paradigm is not only a shift in how we work with LLMs but is also perceived as a shift in how we develop software and handle data. Software and data will no longer be interfaced using user interfaces (UIs), application programming interfaces (APIs), and specialized query languages such as SQL. Instead, they will be designed to be interfaced using natural language.